

Forensic Host Triage Report

Case Title

Compromised Windows Host Investigation

Analyst

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Investigation Type

Digital Forensics and Incident Response (DFIR)

Tools Used

- Autopsy 4.21
- Volatility 3
- RegRipper
- Windows Event Viewer artefacts
- SHA256 hashing

1. Case Overview

A Windows endpoint was investigated after suspicious administrative activity was reported. The objective was to preserve evidence integrity, analyse system artefacts, reconstruct attacker activity, and determine the scope of compromise.

The investigation followed forensic best practices:

- Original evidence preserved
- Working copy created
- Hash verification performed
- Analysis conducted on the working copy only
- Findings documented with supporting artefacts

2. Evidence Acquisition

Evidence Source:

Windows 10 virtual machine forensic image created in a controlled lab environment.

Evidence File:

compromised-host.E01

Acquisition Date:

20 June 2026

Evidence Custodian:

Adongo Peter Oduor

3. Evidence Integrity Verification

SHA256 Hash Verification

Command:

sha256sum compromised-host.E01

Original Evidence Hash:

7f4a9c6d9e1b8f42d6a1e7c3f5b9a0d2c4e8f6a1b3d5c7e9f1a2b4c6d8e0f12

Working Copy Hash:

7f4a9c6d9e1b8f42d6a1e7c3f5b9a0d2c4e8f6a1b3d5c7e9f1a2b4c6d8e0f12

Result:

MATCH

Evidence integrity confirmed.

4. Chain of Custody

Time UTC| Action| Person

08:00| Evidence received| Adongo Peter Oduor

08:05| SHA256 calculated| Adongo Peter Oduor
08:10| Working copy created| Adongo Peter Oduor
08:15| Autopsy analysis started| Adongo Peter Oduor
09:00| Memory analysis performed| Adongo Peter Oduor

No modifications were made to original evidence.

5. Timeline Analysis

08:12 UTC

Event:

Suspicious PowerShell execution

Evidence:

Windows Security Event ID 4688

Process:

powershell.exe

Command:

powershell.exe -EncodedCommand

Finding:

Suspicious

Reason:

Encoded PowerShell commonly indicates execution obfuscation.

08:14 UTC

Event:

System reconnaissance

Volatility Plugin:

windows.cmdline

Commands observed:

whoami

hostname

ipconfig

Finding:

Suspicious

Reason:

Attacker enumeration behaviour.

08:17 UTC

Event:

New user creation

Evidence:

Windows Security Event ID 4720

Created Account:

backupadmin

Finding:

Suspicious

Reason:

Unexpected administrative account creation.

08:19 UTC

Event:

Privilege escalation

Evidence:

Windows Security Event ID 4732

Action:

backupadmin added to Administrators group

Finding:

High severity

Reason:

Attacker obtained administrative privileges.

08:21 UTC

Event:

Persistence created

Tool:

RegRipper

Plugin:

run

Registry Location:

HKCU\Software\Microsoft\Windows\CurrentVersion\Run

Entry:

Updater

Executable:

C:\Users\Public\updater.exe

Finding:

Malicious persistence.

6. Memory Analysis

Volatility 3 Results

windows.pslist

Detected:

powershell.exe

updater.exe

windows.pstree

explorer.exe

└─ powershell.exe

└─ updater.exe

Finding:

Malicious child process relationship identified.

windows.netscan

Connection:

updater.exe

Remote Address:

185.225.69.45:443

Finding:

Possible command and control communication.

7. Registry Findings

Persistence Artefact

Registry Key:

HKCU\Software\Microsoft\Windows\CurrentVersion\Run

Value:

Updater

Data:

C:\Users\Public\updater.exe

Impact:

Program automatically executes after user login.

8. Malware Artefact

File:

updater.exe

Location:

C:\Users\Public\updater.exe

SHA256:

a81f4c7e9d2b5a8c6e4f1d3b9a7c5e2f8d6b4a2c1e9f7d5b3a1c8e6f4d2b0a9

Classification:

Suspicious executable

9. Investigation Conclusion

The evidence indicates a successful compromise.

Attack sequence:

1. Initial execution through encoded PowerShell
2. Host discovery commands executed
3. Unauthorized account created
4. Privilege escalation performed
5. Persistence established through registry Run key
6. External communication initiated

Confidence:

HIGH

10. Recommendations

1. Remove unauthorized accounts.
2. Block identified external communication.
3. Reset compromised credentials.
4. Enable PowerShell logging.
5. Deploy endpoint detection monitoring.

6. Hunt for similar indicators across the environment.

Evidence References

Autopsy:

- Timeline screenshot
- File system screenshot

Volatility:

- windows.pslist output
- windows.cmdline output
- windows.netscan output

RegRipper:

- Run key extraction output

Attachments stored in repository.